Strategies of digestion: digestive efficiency and retention time of forage diets in montane ungulates

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We tested the hypotheses that small-bodied ruminant browsers have higher weight-specific dry matter intake rates, shorter retention times, and lower digestive efficiency than larger-bodied grazers, and that retention time is inversely related to dietary browse concentration. We compared digestive functions of mule deer (*Odocoileus hemionus*), mountain sheep (*Ovis canadensis*), and elk (*Cervus elaphus*) consuming grass–browse diets and related our comparisons to previously reported diet choices of these species. Consistent with our hypotheses, mule deer had higher intake rates, shorter particulate retention times, and reduced digestibilities relative to mountain sheep and elk. Contrary to our hypothesis, increased dietary browse levels increased retention times in all species. Lignification appeared to render forage particles more resistant to comminution rather than less resistant, and consequently increased retention time rather than decreasing it. Browse prolonged retention time of grass, which in turn resulted in elevated fiber digestion in the grass component of the diet. In contrast to mountain sheep and elk, deer were able to compensate for increased retention time and reduced digestibility of browse diets by increasing gut fill, which allowed constant intake of digestible energy despite declining dry matter digestibility. Our findings suggest highly dynamic relationships among intake, fill, and passage, relationships influenced by physiological and morphological traits of the animal and by the fiber composition of the diet.

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Nous avons éprouvé l'hypothèse selon laquelle, chez les brouteurs ruminants de petite taille, les taux de consommation de matières sèches spécifiques à la masse sont plus élevés, la durée de la rétention est plus courte et l'efficacité de la digestion est plus faible que chez les râcleurs de grande taille, hypothèse selon laquelle aussi la durée de la rétention est inversement proportionnelle à la concentration de jeunes pousses dans le régime. Nous avons comparé la digestion chez des Cerfs mulets (*Odocoileus hemionus*), des Mouflons d'Amérique (*Ovis canadensis*) et des Wapitis (*Cervus elaphus*) consommateurs d'herbacées et de jeunes pousses et relié les résultats à ceux d'études sur les choix de régimes de ces espèces. Comme le permettait de croire l'hypothèse, chez les Cerfs mulets, les taux de consommation sont plus élevés, la durée de rétention des particules est plus courte et la digestibilité est plus faible que chez les Mouflons et les Wapitis. Contrairement à l'hypothèse, cependant, une concentration plus élevée de jeunes pousses dans le régime augmente la durée de rétention chez toutes les espèces. La concentration plus grande de lignine semble rendre les particules de nourriture plus résistantes à la transformation plutôt que moins résistantes, ce qui a pour effet d'augmenter la durée de la rétention plutôt que de la diminuer. La présence de jeunes pousses prolonge la durée de rétention des herbacées, ce qui a pour conséquence de rendre plus efficace la digestion des fibres de la composante herbacée du régime. Contrairement aux Mouflons et aux Wapitis, les Cerfs mulets sont capables de compenser l'augmentation de durée de rétention et la diminution de digestibilité des régimes de jeunes pousses par augmentation de la quantité contenue dans le tube digestif, ce qui permet l'absorption constante d'énergie digestible en dépit de la diminution de digestibilité des matières sèches. Nos résultats indiquent qu'il existe des relations très dynamiques entre la consommation, le remplissage et le passage, relations influencées par les caractéristiques physiologiques et morphologiques de l'animal et par la composition en fibres du régime alimentaire.

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Introduction

Relationships among energy requirements, gut capacity, and body mass impose important constraints on diet selection and digestion in wild ruminants. As the body size of ruminants decreases, weight-specific metabolic rate increases while weight-specific gut capacity remains constant (Demment and Van Soest 1985). Consequently, theory suggests that small-bodied ruminants face two nutritional alternatives: they can increase digestive efficiency by choosing foods that are rapidly and completely digested (Hungate *et al.* 1959; Bell 1970; Jarman 1974) or they can increase dry matter intake by choosing foods that are poorly digested, but rapidly excreted (reviewed by Kay *et al.* 1980; Sibly 1981; Hanley 1982). Although studies of ruminant food habits and digestive anatomy lend support to this theory (Hoffman 1973), comparisons of digestive function are essential to its test.

We observed that mule deer (*Odocoileus hemionus*) selected browse-dominated diets substantially less digestible and more lignified than diets chosen by larger-bodied mountain sheep (*Ovis canadensis*) and elk (*Cervus elaphus*) (Hobbs *et al.* 1983; Hobbs and Spowart 1984). These observations, coupled with our review of theory, led to several predictions. We hypothesized that mule deer compensate for reduced digestibility of dry matter by increasing intake, and that the mechanism for this compensation is accelerated turnover of indigestible ingesta (Hobbs *et al.* 1983: p. 10). Thus, from a comparative standpoint, we predicted that mule deer have higher weight-specific intake rates, briefer retention times, and lower digestive efficiency than mountain sheep and elk. Moreover, following the reasoning of Milchunas *et al.* (1978: p. 28) and Van Soest (1982: p. 35), we hypothesized that increasing dietary lignin accelerates rate of passage in montane ungulates by rendering forage particles more susceptible to breakdown during mastication. Here, we report tests of these hypotheses.

Methods

We examined influences of animal species and diet on processes of digestion and excretion in a 3 × 3 factorial experiment with unbalanced

TABLE 1. Chemical composition (%) and gross energy of grass–browse diets fed to mule deer, mountain sheep, and elk (100% dry-matter basis)

	0% browse/ 100% grass	50% browse/ 50% grass	75% browse/ 25% grass
Dry matter	91.4	90.3	89.6
Organic matter	86.0	87.7	88.9
Neutral detergent fiber	69.5	67.4	65.3
Acid detergent fiber	41.3	42.0	42.8
Lignin	4.5	12.2	17.7
Crude protein	6.1	6.3	6.8
Gross energy, kcal/g	4.3	4.6	4.8

replications. Six adult mule deer ($\bar{x}$ weight = 60 kg), three adult mountain sheep ($\bar{x}$ weight = 90 kg), and six adult elk ($\bar{x}$ weight = 250 kg) previously habituated to experimental procedures were offered three forage diets (Table 1). Diet 0 consisted of leaves and stems of overmature grass hay (*Bromus inermis*); diet 50/50 included equal parts of browse stems (*Vaccinium* sp.) and grass hay; diet 75/25 consisted of three parts browse to one part grass hay. Forages were harvested in late summer, air dried, chopped in a silage chopper (1–2 cm), and stored in burlap bags until fed to the animals. We chose these forages because they frequently occur in diets of montane ungulates (Regelin *et al.* 1974; Hobbs *et al.* 1981; Baker and Hobbs 1982; Spowart and Hobbs 1985) and because their chemical composition was particularly well suited to our experimental design. We wished to isolate the effect of lignin in our experiments; consequently, we required diets containing similar concentrations of neutral detergent fiber (NDF) but increasing levels of lignin (Table 1).

Intake, digestion, and passage measurements were made at the Colorado Division of Wildlife Foothills Research Laboratory, Colorado State University, Fort Collins, Colorado. We conducted three experiments with mule deer and elk from October 1982 to April 1983 and three additional ones with mountain sheep from October 1983 to February 1984.

Each experiment consisted of four consecutive phases: adaptation, measurement of ad libitum intake, total fecal and urine collection, and recovery. On day 1, we weighed animals (±0.5 kg) and moved them from 2-ha pastures to individual isolation pens (3 × 10 m). During the subsequent 7 days, animals were gradually switched from their maintenance ration (alfalfa hay + concentrate) to the test diet. Amount of food offered was adjusted until 10–15% of the offering remained uneaten. We divided this amount into two feedings per day (07:00 and 17:00) and measured ad libitum intake for the following 10 days. Animals were then moved to individual digestion cages (3 × 3 m) and fed 95% of the previously determined ad libitum intake. All urine and feces were collected during the following 7 days. Animals were subsequently reweighed, released into pastures, and fed alfalfa and concentrate for 3 weeks before we initiated another experiment.

Three rare earth elements[1] were used as solid phase markers to estimate particulate retention times. To examine effects of animal species, we used ytterbium ($YbCl_3$) as a marker on all diets (0% browse, 7.3 mg Yb/g; 50% browse, 4.5 mg Yb/g; and 75% browse, 4.4 mg Yb/g). To compare turnover times of different diet components across species, we used cerium ($CeCl_3$) to mark the grass portion (9.0 mg Ce/g) and samarium ($SmCl_3$) to mark the browse (7.9 mg Sm/g). These marker combinations allowed us to simultaneously track the passage of grass and browse in each animal and also compare retention times of the entire diet among different ruminant species.

We marked diets immediately before each trial (Teeter *et al.* 1984). Approximately 100–150 g of each diet and (or) diet component was immersed in a 2-L (0.15 *M*) solution of rare earth element and distilled water for 48 h. Rare earth solutions were adjusted to pH 6 to minimize marker migration in the rumen. This mixture was then strained through four layers of cheesecloth, washed six times with distilled water, and dried at 55°C.

On the 1st day of the collection trial, we offered each animal a 20-g dose of the marked forages immediately prior to the morning meal and recorded time of consumption. Any refused portion of the marked forage was collected, weighed, and analyzed to determine intake of each marker. Fecal samples were collected, weighed, composited, and frozen immediately before dosing and at 6, 10, 12, 14, 16, 18, 24, 28, 36, 42, 48, 54, 60, 72, 96, 120, 148, and 168 h after dosing. Because grab sampling was not possible, we used the midpoint of each collection period as the time of defecation. Feed, feces, and refused feed were dried (55°C), ground (1 mm), and analyzed for dry matter (DM), ash, crude protein (Kjeldahl nitrogen × 6.25), neutral detergent fiber (NDF), acid detergent fiber, and acid detergent lignin (Association of Official Analytical Chemists. 1980). Gross energy was determined by bomb calorimetry.

The concentration of stable elements of Yb, Ce, and Sm in feed and fecal samples was determined simultaneously by neutron activation analysis (Gray and Vogt 1974). We estimated rumen and total mean retention time (MRT) of indigestible solids by fitting a two-compartment, time-dependent model (Grovum and Williams 1973) to fecal excretion data (Brandt and Thacker 1958). Fecal marker concentrations were transformed to natural logarithms and regressed on time. Predictions derived from alternative models (Ellis *et al.* 1984) did not differ ($P > 0.10$) from our estimates; however, those models suffered from higher levels of interanimal variability in parameter estimates.

We used the occupancy principle of Steele (1971) to estimate the mass of gut contents (F) as the sum of undigested fill (V_N) and digestible fill (V_D). Undigested fill was estimated as the product of fecal output, FO (g/h), and MRT (h): We estimated digestible fill as follows.

$$V_N = \mathrm{FO} \times \mathrm{MRT}$$

Total intake (I_T) was calculated as $I_T = V_N/(1 - A)$ and digestible intake as $I_D = I_T \cdot A$, where A is the fractional digestibility of the diet. We then estimated digestible fill as the average amount of digestible contents within the tract: $V_D = (I_D \text{ entering tract} - I_D \text{ exiting tract})/2 = I_D/2$.

We tested for treatment main effects and their interaction using a two-way multivariate analysis of variance (Morrison 1976). To account for repeated measures, each vector in the analysis consisted of the responses of an individual animal on the three diets. We compared individual species and diet means with the Bonferroni inequality (Miller 1966) at a simultaneous confidence level of $P = 0.05$. We examined associative effects of increasing browse on NDF digestion by testing for quadratic effects in NDF digestion as a function of diet browse concentration.

Results

Mule deer tended to digest dry (DM) matter less completely than did mountain sheep or elk, but differences among species were significant only for the grass diet (Fig. 1A). Browse decreased dry matter digestion in all species (diet $P = 0.0004$); however, increasing browse levels to 75% had little additional effect. Digestibility of NDF (Fig. 1B) resembled that of dry matter except that deer consistently digested NDF less efficiently than did the other species (species $P = 0.02$, species × diet $P = 0.19$). Increasing dietary browse reduced NDF digestibility in all species (diet $P = 0.0005$).

Consistent with our hypothesis, weight-specific rates of dry matter intake of mule deer exceeded those rates in elk and mountain sheep (species $P = 0.012$, Fig. 1C). Species effects depended on diet (diet × species $P = 0.008$); increasing levels of dietary browse widened differences in intake rates by increasing those rates in mule deer, and decreasing them in elk and mountain sheep. Increased rates of dry matter intake enabled deer to maintain a relatively constant intake of digestible energy (DE) as dietary browse content increased

[1]Research chemicals, P.O. Box 14588, Phoenix, AZ 85063.

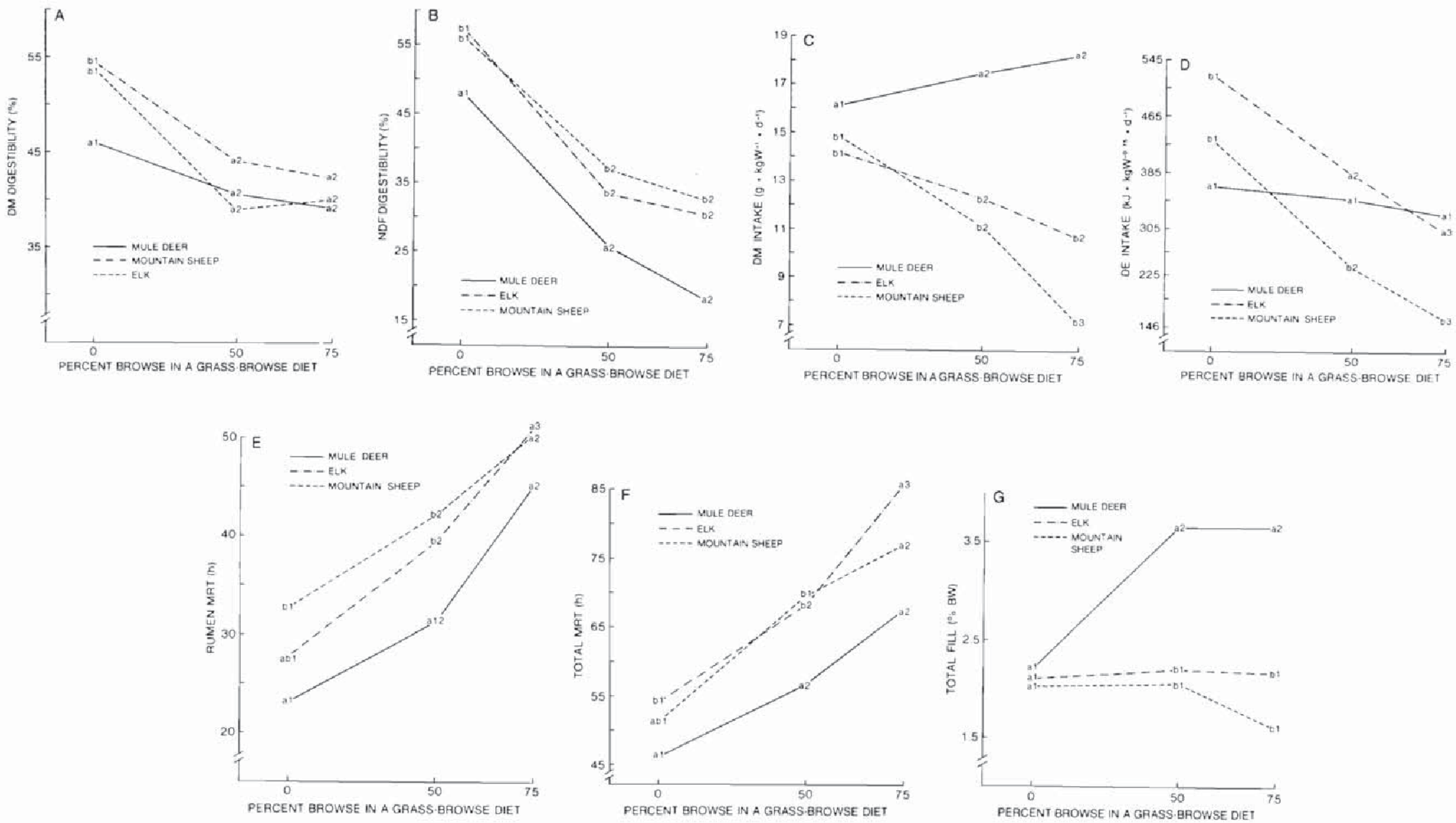

FIG. 1. Apparent digestibility, intake, mean retention time, and dry matter fill in mule deer, mountain sheep, and elk fed grass–browse diets. Different lower case letters indicate significant differences ($P = 0.05$) among species within diets; different numbers indicate differences among diets within species.

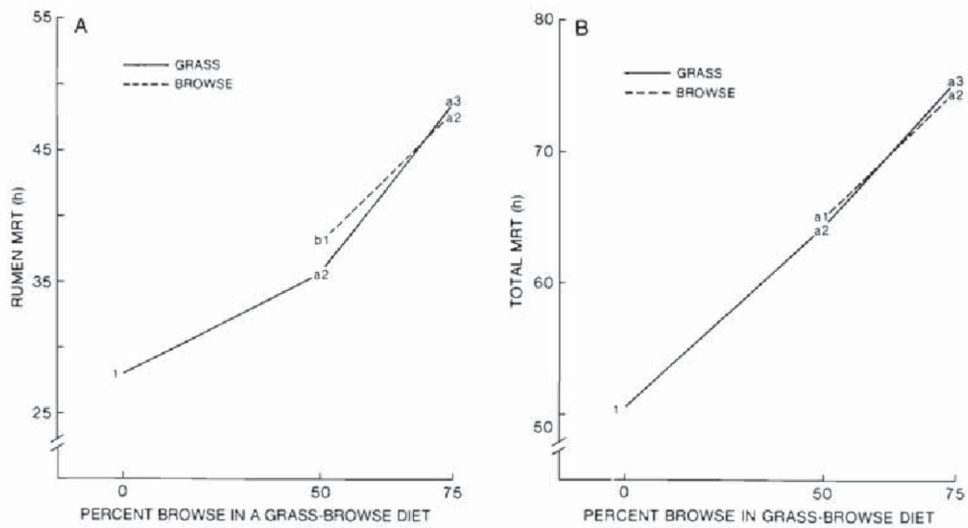

FIG. 2. Influence of increasing browse levels on rumen and total tract retention time of grass and browse. Different letters indicate significant differences ($P = 0.05$) between diet components; different numbers indicate differences among diets. Means represent weighted averages across all animals of all species.

(Fig. 1D) despite reductions in dry matter digestibility (Fig. 1A). In contrast, digestible energy intake in elk and mountain sheep paralleled trends in dry matter digestibility, decreasing sharply with elevated dietary browse levels (species × diet $P = 0.005$).

Contrary to our hypothesis, the increases in dry matter intake by mule deer that we observed in response to increasing dietary browse were not mediated by reduced MRT of ingesta. In all species, MRT in the rumen consistently increased in response to increasing browse (diet $P = 0.031$, species × diet $P = 0.21$, Fig. 1E). We observed parallel trends in total MRT (diet $P = 0.012$, diet × species $P = 0.17$, Fig. 1F). However, differences among species in dry matter intake appeared to be related, at least in part, to interspecific differences in MRT; mule deer consistently showed the shortest MRT (species $P = 0.03$), and the greatest weight-specific intakes.

Increasing intake of dry matter in the face of increasing MRT and decreasing dry matter digestibility is possible only if gut fill increases. Total tract fill in mule deer increased as dietary browse increased, while fill in mountain sheep and elk remained unchanged or declined (Fig. 1G). Weight-specific fill of dry matter (% of body weight) in mule deer tended to exceed fill levels in mountain sheep and elk.

Mean retention time of grass and browse in the rumen and total tract depended on the amount of browse in the diet (Figs. 2A, 2B). We observed small differences in MRT between Ce-marked grass and Sm-marked browse within diets, but saw large differences among diets ($P > 0.05$). We interpret this result to mean that rates of passage of grass and browse components were not independent, that grass and browse passed through the rumen and total tract at essentially equal rates, and that those rates declined as the amount of browse in the diet increased.

We observed significant quadratic effects of diet browse concentration on NDF digestion in elk and mountain sheep ($P = 0.04$) but not in deer ($P = 0.38$). This associative effect appeared to result from effects of increasing browse on grass retention time. Digestibility of NDF in the 0 and 50/50 browse diets was correlated with rumen MRT (Table 2).

TABLE 2. Regressions describing the relationship between NDF digestion (y, %) and rumen MRT (x, h)

	Equation	r	P	s
0% browse/100% grass	$y = 31{:}8 + 0.795x$	63	0.00001	3.2
50% browse/50% grass	$y = 3.54 + 0.784x$	61	0.001	3.8
75% browse/25% grass	NS[a]			

[a]Not significant.

Discussion

In previous experiments (Hobbs *et al.* 1983) we classified mule deer, elk, and mountain sheep as intermediate feeders with elk tending toward the feeding style of grazers, and mule deer and mountain sheep toward concentrate selectors (Hoffman 1973). A fundamental principle of theory on the nutritional ecology of herbivores is that diet choices are related to digestive capabilities and that food habits reflect an animal's physiological and anatomical adaptations to assimilate nutrients. In particular, small-bodied ruminants are believed to be adapted for rapid fermentation and absorption of fermentation products (reviewed by Kay *et al.* 1980; Van Soest 1982: pp. 325–344; Van Hoven and Boomker 1985), thereby allowing them to meet relatively high weight-specific requirements for energy (Demment and Van Soest 1985). However, the success of a strategy of rapid fermentation depends on rapid excretion of indigestible residues, particularly when rate of digestion is high but extent of digestion is low. Otherwise, bulk limitation by the indigestible fraction would limit dry matter intake (Baumgardt 1970; Ammann *et al.* 1973) and negate the benefit of rapid fermentation. Thus small-bodied ruminants, particularly browsers, should be adapted for rapid excretion, as well as rapid digestion.

Patterns of excretion and digestion in mule deer are consistent with the food selection of a small-bodied browser. Regardless of diet, mule deer excreted ingesta more rapidly, and digested fiber less thoroughly, than mountain sheep and elk did. Thus, digestive capabilities of mule deer are compatible with their natural diets, high in cell solubles and low in fiber (Hobbs *et al.* 1983). The superior ability of elk to digest NDF resembles outcomes of other comparisons (Mould and Robbins 1982; Baker and Hansen 1985), and is consonant with our observations of elk diets dominated by graminoids with highly digestible cell walls (Hobbs *et al.* 1983). However, the relatively efficient fiber digestion we observed in mountain sheep (Fig. 1B) appears inconsistent with our previous report of high levels of forbs and cell solubles in their diets (Hobbs *et al.* 1983). This inconsistency can be resolved by changes in mountain sheep diets that occurred as animals matured. Hobbs *et al.* (1983) observed diets of mountain sheep lambs. As adults, those same animals chose predominantly grass diets containing high levels of NDF and digestible dry matter (Dailey *et al.* 1984; Hobbs and Spowart 1984; Spowart and Hobbs 1985), diet choices compatible with efficient digestion and relatively long retention of fiber observed in this study.

Milchunas *et al.* (1978: p. 28) and Van Soest (1982: p. 35) suggested that lignin may make plant cell walls brittle, thereby accelerating comminution of forage particles during mastication and rumination. This, in turn, would lead to decreased retention time by increasing the likelihood of particle escape from the rumen. Previous comparisons of retention times of forages differing in lignin content have frequently been confounded by differences in NDF concentration (Milne and Bagley 1976; Milchunas *et al.* 1978; Poppi *et al.* 1980; Foose 1982). However, increasing browse levels in our test diets increased dietary lignin concentration without markedly changing the NDF concentration. Thus, we can unambiguously reject the hypothesis that increasing lignin accelerates rate of passage, a conclusion similar to that of Spalinger *et al.* (1986) who observed negative correlations between lignin concentration of forages and their breakdown rates in rumens of mule deer and elk. It follows that lignification makes forage particles more resistant to comminution rather than more susceptible, and consequently increases retention time rather than decreasing it. We concur with the argument of Spalinger *et al.* (1986) that it is the low cell-wall content of forbs and browse leaves, rather than their high lignin concentrations, that allows their rapid excretion and makes them suitable food for small-bodied ruminants.

Mean retention times of browse and grass increased with the proportion of browse in the diets. We surmise that browse influenced the passage of grass, rather than vice versa, because we observed large differences in MRT (rumen and total) for Ce-marked grass between the 0 and 50/50 diets and small differences in MRT between Ce-marked grass and Sm-marked browse within the 50/50 diet. Because grass appeared to take on the rate of passage of browse, we suggest that grass particles may have become physically entrapped in the mass of browse ingesta.

The increase in MRT of grass and browse between the 50/50 and 75/25 diet is more difficult to explain. Apparently, increasing dietary browse increased retention of browse as well as grass because *both* components showed a longer MRT on the 75/25 diet than on the 50/50. Increasing levels of dietary fiber and consequent increased resistance to particle breakdown can be compensated for by accelerated rumination time (Bines and Davey 1970; Welch and Smith 1970; Welch 1982; Murphy *et al.* 1983). However, that compensation is limited; cattle and sheep will not ruminate more than 8–12 h/day regardless of dietary fiber levels (Welch 1982). Based on the relationship between cell wall intake and rumination time synthesized by Robbins (1983: Fig. 14.5), we infer that elk, mountain sheep, and mule deer were ruminating maximally on the 0 browse diet, and consequently could not increase rumination time as dietary browse increased. Thus, assuming that lignification increased resistance of forage particles to breakdown (Spalinger *et al.* 1986), increasing levels of browse provided more recalcitrant particles at the same level of rumination, and consequently reduced rate of comminution and passage of the diet.

Our values for MRT resemble other estimates for roughage diets in ruminants (reviewed by Warner 1981). Associative effects on MRT have been seen before (Bines and Davey 1970; Ledoux *et al.* 1985); however, Milne and Bagley (1976) offered the only previous report of interactions in MRT between low quality forages. Similar to our findings, they also found that increasing levels of browse in a grass–browse diet fed to sheep increased browse MRT in the total tract. Although MRT of grasses also tended to accelerate, that tendency was not significant ($P > 0.05$) (Milne and Bagley 1976). In contrast to our results, MRT of grasses in their experiment was substantially lower than browse MRT. However, Milne and Bagley (1976) fed diets containing lower NDF levels than ours (45 vs. 70%), and they offered grass and browse separately so that the amount of browse in the diet varied according to the choice of the sheep. If the browse and grass components were eaten sequentially, rather than together as in our study, then the opportunity for associative effects was reduced relative to our experiments. Despite these differences, the similarities between our findings and those of Milne and Bagley (1976) offer an important message: the retention of individual forages in the gut of ruminants can be strongly associative.

Moreover, our results compel the conclusion that associative effects on MRT will likely have associative effects on fiber digestion. The nonlinearity in relationships between dietary browse levels and NDF digestion in elk and mountain sheep indicates that NDF digestibility was not simply a weighted average of the digestibilities of NDF in diet components, but instead resulted from the interactions of those components. We surmise that browse increased MRT of grass, which in turn resulted in elevated NDF digestion of the grass component. Thus, the "handling times" (Spalinger *et al.* 1986) and digestibilites of forages can depend on the collective properties of diet mixtures as well as the individual properties of diet constituents. This result has fundamental implications for modeling diet selection and digestion.

Variation in rates of dry matter intake in ruminants given ad libitum forage depends on rates of dry digestion, excretion, and the capacity of rumen or total tract. That is, increased rate of intake can be accommodated by increased rates of ruminal emptying via absorption or escape, or by increased fill (reviewed by Van Soest 1982: pp. 281–290; Ellis *et al.* 1984; Grovum 1984). Gut fill has been shown to vary in response to changes in diet composition and energy demands in ruminants (Tulloh and Hughes 1965; Tulloh 1966; Journet and Remond 1976; Milne *et al.* 1978; Hartnell and Satter 1979; Staaland *et al.* 1979) and other mammals (Bohman *et al.* 1953; El-Harith *et al.* 1976; Gross *et al.* 1985). Increases in gut fill, rather than accelerated rate of passage, allowed mule deer to increase intake in response to increasing dietary browse levels. Although this response may be due to differences in palatability of diets among species, we suggest that the failure of elk and mountain sheep to increase gut fill follows from relationships between rumen volume and fill. The volume of ingesta in small-bodied concentrate selectors like mule deer normally occupies only a portion of their available rumen space, thus allowing them to increase intake by increasing fill when forage quality declines. In contrast, rumens of larger-bodied grazers tend to be filled to capacity (Short *et al.*

1965; Hoffman 1973; Bunnell and Gillingham 1985: Fig. 3). It is also possible that the lumen of the digestive tract expands through elaboration of gut tissue, but such changes are energetically expensive (Koong *et al.* 1983; Smith and Baldwin 1974; Gross *et al.* 1985). In modeling forage intake and turnover in relation to body size, Demment and Van Soest (1985) treated fill as a variable influenced by animal species, but not by diet. Our findings suggest that their view oversimplifies a highly dynamic relationship among intake, fill, and turnover, a relationship that is influenced by species *and* diet.

Our findings support the tenet of nutritional ecology that processes of diet selection are closely linked to those of digestion, and that diet choices reflect the physiological and anatomical adaptations of the choosers. However, characteristics of forages also constrain food habits; in particular, we surmise that lignified diets chosen by small-bodied browsers like mule deer must contain low levels of cell wall to allow the rapid excretion necessary for their digestive strategy. Regardless of diet lignin levels, larger-bodied ruminants such as elk and mountain sheep appear better able to digest dietary cell wall than smaller ruminants such as mule deer.

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